

Mathematical Analysis Lecture 22 积分学.

e.g. $\int \arccos x \, dx = x \cdot \arccos x - \int x \cdot d(\arccos x) = x \cdot \arccos x + \int \frac{x}{\sqrt{1-x^2}} \, dx = x \cdot \arccos x - \sqrt{1-x^2} + C.$

e.g. $\int \frac{\ln \cos x}{\cos^2 x} \, dx = \ln \cos x \cdot \tan x - \int \tan x \, d(\ln \cos x) = \ln \cos x \cdot \tan x + \int \tan^2 x \, dx = \ln \cos x \cdot \tan x + \int (\sec^2 x - 1) \, dx$
 $= \ln \cos x \cdot \tan x + \tan x - x + C.$ Note: $\tan^2 x = \sec^2 x - 1$ 似乎是一个相当常用的代换.

e.g. $\int e^{\sqrt{x}} \, dx$ 设 $\sqrt{x} = t \therefore x = t^2 \therefore$ 原 $= \int e^t dt^2 = 2 \int t e^t \, dt = 2 t e^t - \int e^t \, dt = 2 t e^t - e^t + C = 2 \sqrt{x} e^{\sqrt{x}} - e^{\sqrt{x}} + C$

e.g. $\int \sqrt{x^2+a^2} \, dx = x \sqrt{x^2+a^2} - \int x \, d\sqrt{x^2+a^2} = x \sqrt{x^2+a^2} - \int \frac{x^2}{\sqrt{x^2+a^2}} \, dx = x \sqrt{x^2+a^2} - \int (\sqrt{x^2+a^2}) \, dx + \int \frac{a^2}{\sqrt{x^2+a^2}} \, dx.$

$\therefore 2 \int \sqrt{x^2+a^2} \, dx = x \sqrt{x^2+a^2} + a^2 \ln(x + \sqrt{x^2+a^2}) + C.$

e.g. 求 $\int \frac{dx}{(x^2+a^2)^n}$ $= x \cdot \frac{1}{(x^2+a^2)^n} - \int x \, d((x^2+a^2)^{-n}) = x \cdot \frac{1}{(x^2+a^2)^n} - \int x \cdot (-n) \cdot (x^2+a^2)^{-(n+1)} \cdot (2x) \, dx$

设 $I_n = \int \frac{dx}{(x^2+a^2)^n}$ $\therefore I_{n+1} = \int \frac{dx}{(x^2+a^2)^{n+1}}$ 原 $= I_n = \frac{x}{(x^2+a^2)^n} + 2n \int \frac{dx}{(x^2+a^2)^n} - 2a^2 n \int \frac{dx}{(x^2+a^2)^{n+1}}$

$\therefore I_{n+1} = \frac{1}{2na^2} \frac{x}{(x^2+a^2)^n} + \frac{2n-1}{2na^2} I_n$ 找到递推公式即可

e.g. 设 $I_n = \int \sec^n x \, dx$, 求证: $I_n = \frac{1}{n-1} \sec^{n-2} x \cdot \tan x + \frac{n-2}{n-1} I_{n-2}$

$I_n = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-3} x \sec x \cdot \tan^2 x \, dx$

$I_n = \sec^{n-2} x \cdot \tan x - (n-2) I_n + (n-2) I_{n-2}$ QED

有理函数的积分: 任何有理函数都可积

定义: 有理函数: $R(x) = \frac{P(x)}{Q(x)} = \frac{a_0 x^n + a_1 x^{n-1} + \dots + a_n}{b_0 x^n + b_1 x^{n-1} + \dots + b_n}$ 称为有理函数

$m \leq n$ 时, $R(x)$ 为有理函数. 反之为真分式. 任意有理函数都可以写为多项式与真分式之和.

真分式可拆分为 $\frac{A}{(x-a)^k} + \frac{Mx+N}{(x^2+px+q)^k}$ (小心 $k=1$ 的情况). ($k \in \mathbb{N}^+$, $p^2-4q < 0$)

引理: 首系数为1的实系数多项式 $Q(x) = (x-a)^\alpha (x-b)^\beta \dots (x^2+px+q)^\mu (x^2+rx+s)^\delta \dots$

定理: 若 $R = \frac{P}{Q}$ 是真分式, 且有如上所示的分解, 则有唯一的分解形式:

$R(x) = \sum_{j=1}^{\alpha} \frac{A_j}{(x-a)^j} + \dots + \sum_{j=1}^{\beta} \frac{B_j}{(x-b)^j} + \sum_{j=1}^{\mu} \frac{2K_j x + L_j}{(x^2+px+q)^j} + \dots + \sum_{j=1}^{\delta} \frac{2M_j x + N_j}{(x^2+rx+s)^j}$ 尽管复杂, 但可以积

e.g. $\frac{1}{x(x-1)^2} = \frac{x-(x-1)}{x(x-1)^2} = \frac{1}{(x-1)^2} - \frac{1}{x(x-1)} = \frac{1}{(x-1)^2} - \frac{1}{x-1} + \frac{1}{x}$

e.g. $\frac{x+3}{x^2-5x+6} = \frac{x+3}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3} = \frac{Ax-3A+Bx-2B}{x^2-5x+6} \begin{cases} A+B=1 \\ -3A-2B=3 \end{cases} \Rightarrow \begin{cases} A=-5 \\ B=6 \end{cases}$

Or, 我们有 $A = (x-2) \cdot \text{原式} |_{x=2} = -5$, $B = (x-3) \cdot \text{原式} |_{x=3} = 6$. 特殊类型的待定系数法.

e.g. $\frac{1}{(2x+1)(x^2+1)} = \frac{A}{2x+1} + \frac{Bx+C}{x^2+1}$ $A = (2x+1) \cdot \text{原式} |_{x=-\frac{1}{2}} = \frac{4}{5}$ $\frac{1}{2} x=0 \Rightarrow C = \frac{1}{5}$ $\frac{1}{2} x=1 \Rightarrow B = -\frac{2}{5}$

$\int \frac{A}{x-a} \, dx = A \ln|x-a| + C$ $\int \frac{A}{(x-a)^n} \, dx = \frac{A}{1-n} (x-a)^{1-n} + C.$

$\int \frac{Mx+N}{x^2+px+q} \, dx = \int \frac{\frac{M}{2}(2x+p) + N - \frac{Mp}{2}}{x^2+px+q} \, dx$ 在理论上, 有理函数可积

e.g. $\int \frac{x-2}{x^2+2x+3} \, dx = \int \frac{\frac{1}{2}(2x+2)-3}{x^2+2x+3} \, dx = \frac{1}{2} \ln(x^2+2x+3) - \frac{3}{2} \arctan(\frac{x+1}{\sqrt{2}}) + C$

e.g. $\int \frac{2x^3+2x^2+5x+5}{x^4+5x^2+5} \, dx = \frac{1}{2} \int \frac{d(x^4+5x^2+5)}{x^4+5x^2+5} + \int \frac{(x^2+1)(x^2+4)}{(x^2+1)(x^2+5)} \, dx$

e.g. $\int \frac{dx}{x^4+1} = \frac{1}{2} \int (\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}} - \frac{1-\frac{1}{x^2}}{x^2+\frac{1}{x^2}}) \, dx = \frac{1}{2} \int \frac{d(x-\frac{1}{x})}{(x-\frac{1}{x})^2+2} - \frac{1}{2} \int \frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2-2}$